What Makes a Generalist Robot Policy?

A Study of Models, Tasks, and Data for Low-Cost Bimanual Manipulation

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***Abstract*—Generalist robot policies aim to reuse knowledge across tasks, environments, and embodiments while retaining reliable closed-loop control. This work studies several ingredients of that problem using a low-cost, custom-built fixed-base bimanual robot with 12 actuated degrees of freedom (DoF): each side has five arm-joint DoF and one independently actuated gripper. We compare Action Chunking Transformer (ACT) and the pretrained vision-language-action model $\pi_{0.5}$ under matched robot-data conditions, and perform parallel within-family ablations of simulated data and multi-task training. For each model family, we compare real-only and real-plus-simulation training, seven independently trained single-task specialists and one jointly trained seven-task policy, and robustness under standardized physical perturbations. Because raw human video does not provide robot joint actions, the egocentric-human-data ablation is performed only for $\pi_{0.5}$ using a cross-embodiment end-effector representation derived from RGB-D hand tracking. We evaluate all configurations through repeated closed-loop rollouts under matched initial conditions and report complete success, partial success, inference latency, and 95% confidence intervals. Recovery is evaluated separately with controlled physical perturbations. The scope of the study is deliberately narrow: conclusions are restricted to the custom robot embodiment, the seven benchmark tasks, and the available data and compute budget. The robot's 3D-printable parts, MuJoCo simulation environment, and model training and inference code are available at https://github.com/tmeynier/bimanual-models-tasks-data.**

***Index Terms*—bimanual manipulation, robot learning, vision-language-action models, behavior cloning, simulation, egocentric data, closed-loop evaluation**

I. Introduction

Recent robot-learning research has increasingly shifted from policies designed for one robot and one task toward *generalist robot policies*: models that can reuse knowledge across tasks, environments, objects, and embodiments. This direction is inspired by the foundation-model paradigm in language and vision: training increasingly capable models on larger and more diverse datasets, using greater computational resources, generally improves performance and produces representations that can be adapted to many downstream tasks. Robot learning aims to follow the same paradigm, but presents additional challenges. A robot policy must generate continuous, temporally correlated actions under partial observability, physical contact, sensor noise, and strict latency constraints. Moreover, each prediction changes the state from which the next prediction is made, so small errors can compound into unrecoverable failures. Robot-interaction data are also far more expensive to collect than text or images.

Historically, much of robot learning has treated manipulation as a collection of task-specific problems: a policy is trained for one robot, one task, and one demonstration distribution, without relying on a general-purpose pretrained robot policy. Behavior cloning learns control directly from expert demonstrations, but can suffer from distribution shift when the learned policy visits states not covered by those demonstrations [1]. Action Chunking Transformer (ACT) improves temporal consistency by predicting sequences of future actions and has demonstrated precise bimanual manipulation using low-cost hardware [2]. Generative approaches such as Diffusion Policy instead model *multimodal* distributions over action trajectories, where multi-modality means that several distinct action modes can be valid for the same observation and task [3]. For example, a robot may reach the same object from different directions or use different grasp strategies. Modeling these alternatives as a distribution avoids collapsing them into a single averaged action that may not correspond to any valid behavior. These specialized methods remain strong baselines when the task and operating conditions are well specified, but they represent a narrower paradigm than generalist policies: their semantic knowledge, task coverage, and ability to transfer across embodiments are largely determined by the task-specific robot demonstrations on which they are trained.

More recent vision-language-action (VLA) models follow a broader pretraining paradigm. They adapt pretrained vision-language representations, learned largely from Internet-scale image and text data, to robot control. Models in the π_0 family combine a pretrained vision-language backbone with a continuous flow-matching action expert [5]. Throughout this paper, "π_0 family" refers to this policy lineage, whereas $\pi_{0.5}$ denotes the specific member used in our experiments. In parallel, multi-task and cross-embodiment training increase the diversity of physical experience available to the policy. Open X-Embodiment showed positive transfer from data collected across many robot platforms [6], while $\pi_{0.5}$ demonstrated that heterogeneous robot, semantic, language, and web data can support long-horizon manipulation in previously unseen environments [7]. Together, Internet-scale semantic pretraining and diverse robot experience provide a route toward policies whose knowledge is not tied to a single task or embodiment.

A natural continuation of cross-embodiment learning is to treat humans as an additional embodiment. Egocentric human video is particularly attractive because it can capture large numbers of manipulation tasks, objects, and environments at a scale that is difficult to obtain with physical robots. Kareer et al. [10] show that human-to-robot transfer emerges when a VLA is pretrained on sufficiently diverse robot scenes, tasks, and embodiments, suggesting that large-scale pretraining can produce representations that are increasingly agnostic to the acting embodiment. EgoScale further demonstrates predictable scaling from 20,854 hours of action-labeled egocentric human video and transfers the resulting motor priors to dexterous robot hands with limited robot supervision [11]. These findings make egocentric human experience a promising scalable source of cross-embodiment supervision for robot learning.

Beyond direct policy learning, another emerging direction is the use of *world models*. A world model is a learned predictive model of environment dynamics: given the current state or observation and, in an action-conditioned setting, candidate actions, it predicts how the world is likely to evolve. Such predictions can support planning by allowing an agent to compare possible future outcomes before executing an action. Joint-Embedding Predictive Architectures (JEPAs) learn predictive structure in representation space rather than reconstructing every image pixel [8]. V-JEPA 2 scales this idea to more than one million hours of Internet video, and its action-conditioned variant V-JEPA 2-AC is post-trained on robot trajectories to predict future latent states under candidate actions, enabling model-based planning on physical robots [9]. This family illustrates a complementary path toward general-purpose robot intelligence: learning not only a policy that maps observations to actions, but also an internal predictive model of how actions change the physical world.

This technical report asks how several ingredients commonly associated with more general robot policies behave in a deliberately small, resource-constrained setting. The study uses a custom-built robot embodiment together with behavior-cloning demonstrations, simulated trajectories, and egocentric human data. Across seven tasks, we ask: *what is the incremental effect of simulated experience, multi-task parameter sharing, and egocentric human supervision on closed-loop task success and robustness in bimanual manipulation?* We additionally perform one exploratory zero-shot evaluation on a shape-conditioned sorting task held out from all benchmark fine-tuning data. We treat these as benchmark-specific measurements rather than evidence of general open-world capability.

We study this question using a low-cost, custom-built fixed-base bimanual robot with 12 actuated degrees of freedom (DoF): ten pose-controlling arm joints and two independently actuated grippers. The benchmark contains seven tasks, listed in approximate order of increasing difficulty:

1. **Placement — Basic pick-and-place:** Pick up a cube and place it inside a target container.
2. **Picking — Object retrieval:** Pick up a cube from inside a container and place it outside the container.
3. **Sorting — Color-conditioned sorting:** Sort the blue and green cubes into their respective containers.
4. **Articulated manipulation — Drawer opening:** Interpret a drawer number, locate the corresponding drawer, grasp its handle, and open it.
5. **Precision manipulation — Cube stacking:** Pick up the green cube and place it stably on top of the blue cube.
6. **Bimanual collaboration — Mid-air handover:** Pick up a cube from a container with the right arm, transfer it to the left arm in mid-air, and place it inside a second container.
7. **Long-horizon manipulation — Warehouse stocking:** Retrieve the cube of the requested color, open the specified drawer, place the cube inside it, and close the drawer.

Task durations range from about 10 s for Tasks 1–6 to 20 s for Task 7.

We organize the experiments so that simulation and task-diversity effects are measured within both model families. Comparisons between ACT and $\pi_{0.5}$ remain system-level comparisons rather than architecture-only ablations because the models differ in pretraining, scale, task conditioning, and action parameterization.

- **E1. Single-task model-family comparison:** ACT + real-robot BC and $\pi_{0.5}$ + real-robot BC are trained on Task 1 using the same robot demonstrations and evaluated under matched rollout conditions.
- **E2. Simulation comparisons in both model families:** for ACT and $\pi_{0.5}$ separately, a Task 1 real-only policy is compared with real-plus-simulation training under two protocols. The fixed-budget ablation holds the total number of optimization examples constant; the additive experiment preserves real-data exposure while adding simulated examples and optimization steps.
- **E3. Task-diversity ablation in both model families:** for ACT and $\pi_{0.5}$ separately, seven independently trained single-task specialists are compared with one jointly trained seven-task policy. The same real and simulated datasets are used per task, and the expected number of optimization examples per task is matched between specialist and multi-task conditions.
- **E4. Egocentric-data ablation for $\pi_{0.5}$:** the seven-task $\pi_{0.5}$ policy is trained with or without aligned egocentric human data under the same total optimization budget and real-robot sampling rate. Human batches replace a matched fraction of simulation batches.
- **E5. Held-out-task generalization:** the trained multi-task ACT and $\pi_{0.5}$ policies are evaluated without further training on a shape-conditioned sorting task: place a rectangular block in the left container and a cube in the right container. This task is excluded from all benchmark training and validation sets. The $\pi_{0.5} + \text{Ego}$ policy is reported as an additional exploratory generalization condition.
- **E6. Controlled perturbation evaluation:** the trained multi-

task ACT and $\pi_{0.5}$ policies are evaluated without further training under the same standardized physical interventions to measure recovery separately from nominal task success. The $\pi_{0.5}$ + Ego policy is reported as an additional exploratory recovery condition.

All configurations are evaluated through matched closed-loop rollouts. Complete and partial task success quantify nominal performance, while recovery is measured separately with standardized physical perturbations. Offline action error is used to select a checkpoint for each training run and is also reported as a training diagnostic; it is not treated as the primary performance metric because it does not capture compounding errors or closed-loop recovery.

a) Contributions.:

- We introduce a low-cost, 3D-printed bimanual robot, together with its MuJoCo simulation environment, four-view visual setup, and a seven-task benchmark.
- We provide matched within-family ablations for both ACT and $\pi_{0.5}$, measuring the effect of simulated demonstrations and multi-task parameter sharing while avoiding attribution of ACT–$\pi_{0.5}$ differences to a single architectural factor.
- We study whether RGB-D egocentric human demonstrations can provide additional supervision to $\pi_{0.5}$ through a human-hand/robot-end-effector representation, without fabricating robot joint labels for human video.
- We test generalization without additional fine-tuning on a shape-conditioned sorting task excluded from all benchmark training and validation data.
- We evaluate recovery separately from nominal success using controlled physical perturbations, allowing robustness differences between the trained policies to be measured directly.

II. Related Work

Research toward general-purpose robot manipulation has progressed from task-specific imitation learning to vision-language-action policies, cross-embodiment training, human demonstrations, and predictive world models.

a) Task-specific imitation learning.: Behavior cloning learns robot control directly from demonstrations, but errors can compound when the policy encounters states absent from the training data [1]. ACT addresses long-horizon bimanual manipulation by predicting action chunks and temporally ensembling overlapping predictions [2]. Diffusion Policy and flow-matching methods instead model distributions over action sequences, allowing multiple valid behaviors to be represented [3], [4].

b) Vision-language-action policies.: Vision-language-action (VLA) models connect pretrained visual-language representations to continuous robot control. The original π_0 combines a vision-language backbone with a flow-matching action expert [5]. Its successor, $\pi_{0.5}$, introduces heterogeneous co-training and high-level semantic supervision to improve open-world generalization [7]. We use $\pi_{0.5}$, rather than the original π_0, as the common pretrained base policy in our experiments.

Insert three panels showing: (a) the leader arms used for teleoperation, (b) the physical follower robot, and (c) the MuJoCo digital twin.

Fig. 1. Robot platform and digital twin. The system consists of (a) leader arms used for teleoperation and demonstration collection, (b) the physical fixed-base bimanual follower robot, and (c) the corresponding MuJoCo simulation environment.

c) Cross-embodiment robot learning.: Cross-embodiment learning investigates whether experience collected with one physical system can improve control on another. This requires handling differences in camera viewpoints, kinematics, action spaces, and hardware. Open X-Embodiment showed that combining trajectories from heterogeneous robots can produce positive transfer across platforms [6].

d) Egocentric human experience.: Humans can be viewed as an additional embodiment for which large and diverse datasets are considerably easier to collect than physical robot trajectories. Egocentric video therefore provides a scalable source of manipulation experience, although human-to-robot transfer must bridge differences in morphology, viewpoint, and action representation. Kareer et al. [10] co-fine-tune $\pi_{0.5}$ on robot and egocentric human demonstrations; we denote this formulation as $\pi_{0.5}$ + Ego. EgoScale reports predictable improvements as the amount of action-labeled human video increases [11]. More recently, EgoWAM shows that predicting future DINO feature representations from egocentric data can improve out-of-distribution object and scene generalization by up to four times [14]. These results suggest that human experience may provide a scalable source of cross-embodiment supervision, particularly when the learned representation captures embodiment-invariant scene dynamics rather than human actions alone.

e) Predictive world models.: World models are not evaluated in our ablations, but represent a complementary direction for future work.

Motivated by these large-scale studies, we examine whether some of these trends are observable on a custom, low-cost robot under limited data and compute.

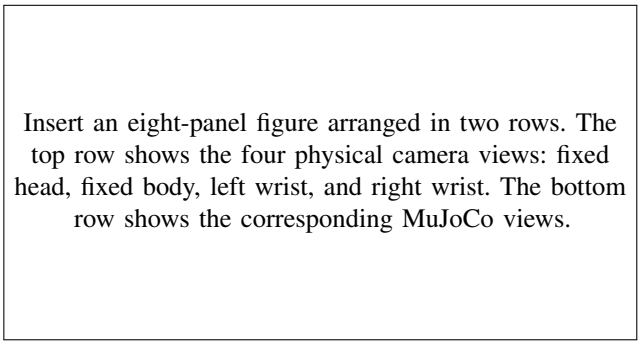


Fig. 2. Real and simulated visual observations. The top row shows the four physical camera views from the fixed head, fixed body, left wrist, and right wrist cameras. The bottom row shows the corresponding MuJoCo viewpoints. The simulated cameras are initialized from hardware-matched nominal poses and slightly randomized around these poses during training.

III. Robot Architecture

The platform is a low-cost, 3D-printed fixed-base bimanual robot with 12 actuated degrees of freedom (DoF):

$$N_{\mathrm{DoF}} = 2 \times (5 \text{ arm-joint DoF} + 1 \text{ gripper DoF}) = 12.$$

The complete system, including the leader arms, physical follower robot, and MuJoCo digital twin, is shown in Fig. 1.

A. Bimanual Manipulation Arms

The robot uses two 3D-printed arms. Each arm is actuated by six Feetech STS3215 serial-bus servomotors, following HuggingFace's LeRobot SO-101 follower-arm configuration [15]. Five joints control the arm configuration and wrist orientation, while the sixth independently actuates the gripper. Thus, each manipulator has five kinematic DoF that determine its end-effector pose plus one gripper DoF. The gripper actuator is included in the 12-DoF system count but does not provide an additional Cartesian pose DoF.

The serial-bus servomotors provide commanded and measured joint positions, allowing the controller to operate the arms using joint-position feedback. The two arms are fixed relative to the workspace; all benchmark manipulation is therefore performed through arm and gripper motion alone. Objects and articulated fixtures are arranged inside the reachable workspace of the bimanual system.

B. Visual Observations

The robot uses four RGB viewpoints, shown in the top row of Fig. 2. An Intel RealSense D415 is mounted on the fixed head above the arms and provides a global frontal view of the workspace. A second 30 fps USB RGB camera is mounted in the fixed robot body below the arms, providing a lower global view. Two additional 30 fps USB RGB cameras are mounted on the left and right wrists and provide close-range views of end-effector–object interactions. During robot training and deployment, the D415 is used in RGB-only mode; its depth stream is not provided to ACT or $\pi_{0.5}$.

The MuJoCo environment reproduces the same four nominal camera placements and view directions as the physical robot, as shown in the bottom row of Fig. 2. Simulated observations are rendered from matched head, body, left-wrist, and right-wrist viewpoints. Simulated camera poses are randomly perturbed around their hardware-matched nominal poses during training to improve robustness to viewpoint variations. Lighting and textures may also be randomized.

Before operation, the robot undergoes a calibration procedure in which the arm joint offsets are established for both arms.

The robot's custom morphology and multi-view observation setup make embodiment adaptation an important part of the study. Particularly this embodiment is not represented in the original $\pi_{0.5}$ training data.

C. Visual Observations

The robot uses four RGB viewpoints. An Intel RealSense D415 is mounted on the fixed head above the arms and provides a global frontal view of the workspace. A second 30 fps USB RGB camera is mounted in the fixed robot body below the arms, providing a lower global view. Two additional 30 fps USB RGB cameras are mounted on the left and right wrists and provide close-range views of end-effector/object interactions. During robot training and deployment, the D415 is used in RGB-only mode: its depth stream is not provided to ACT or $\pi_{0.5}$.

The MuJoCo environment reproduces the same four nominal camera placements and view directions as the physical robot. Simulated training observations are rendered from these matched head, body, left-wrist, and right-wrist viewpoints. Camera extrinsics are therefore held fixed to the corresponding real-hardware mounts in the main experiments; appearance properties such as lighting and textures may still be randomized.

Before operation, the robot undergoes a calibration procedure in which the arm joint offsets are established for both arms. The fixed head/body cameras are extrinsically calibrated to the robot frame, while the wrist-camera transforms are defined relative to their corresponding wrist links.

The robot's custom morphology and multi-view observation setup make embodiment adaptation an important part of the study.

IV. Task Suite

The seven benchmark tasks are summarized in Table I in approximate order of increasing difficulty.

TABLE I
MANIPULATION TASKS, ORDERED BY APPROXIMATE DIFFICULTY. ALL OBJECTS AND FIXTURES ARE POSITIONED WITHIN THE FIXED BIMANUAL WORKSPACE.

Task title	Language instruction
T1: Basic object placement	"Pick up the cube and place it inside the target container."
T2: Object retrieval	"Pick up the cube from the container and place it outside the container."
T3: Color-conditioned sorting	"Sort the blue and green cubes into their respective containers."
T4: Number-conditioned drawer opening	"Open the drawer with the requested number."
T5: Precision cube stacking	"Place the green cube stably on top of the blue cube."
T6: Bimanual mid-air handover	"Pick up the cube from the first container with the right arm, transfer it to the left arm in mid-air, and place it inside the second container."
T7: Warehouse stocking	"Retrieve the cube of the requested color, open the requested drawer, place the cube inside, and close the drawer."

V. STUDY DESIGN

A. Operational definition of a generalist policy

We emphasize that, in this report, "generalist" is used in a restricted benchmark sense. A single policy is considered more general within this benchmark when it can execute a larger portion of the seven-task suite without using a separate model for each task. We additionally report one exploratory held-out shape-sorting test, but this does not establish broad zero-shot generalization, transfer to new environments, or transfer to a new robot embodiment. We report *complete success*, indicating whether the full task is completed, and *partial success*, defined as the fraction of predefined subtasks completed during the longer-horizon task. These metrics do not directly penalize collisions or undesirable execution paths when the task is nevertheless completed, so all rollouts are also inspected for anomalous behavior. Recovery is not inferred from nominal success; it is evaluated separately using the controlled perturbation protocol described in Sec. IX. For the longer-horizon task, partial success is computed from a predefined sequence of task-specific subgoals.

B. Tested Hypotheses

The experiments test the following benchmark-specific hypotheses. We report both positive and negative changes and avoid treating a null result as evidence that an ingredient is unimportant outside this setup.

H1: Model-family comparison
Under the same Task 1 robot demonstration set, ACT and $\pi_{0.5}$ may differ in success and inference latency. Because the systems differ in pretraining, scale, task conditioning, and action parameterization, this comparison is descriptive and is not used to isolate any one factor.

H2: Simulation data
Within each model family, we report both a fixed-budget comparison, in which simulated batches replace repeated real-data batches, and an additive comparison, in which real-data exposure is preserved while simulation and optimization steps are added. The former isolates the data-source substitution under fixed compute; the latter measures the practical effect of adding simulation when additional compute is allowed.

H3: Multi-task parameter sharing
Within each model family, with matched per-task datasets and optimization exposure, a single seven-task policy may perform differently from seven independently trained specialists. A positive difference is interpreted as evidence consistent with beneficial transfer within this benchmark; a negative difference indicates interference.

H4: Egocentric human data
For $\pi_{0.5}$, under a fixed total optimization budget and fixed real-robot sampling rate, replacing a portion of simulation batches with aligned egocentric human data may change multi-task performance. This condition is not mirrored for standard ACT because human videos do not provide the robot joint-action targets expected by that baseline.

H5: Held-out-task generalization
Without additional fine-tuning, the multi-task $\pi_{0.5}$ policy may transfer to a shape-conditioned sorting instruction excluded from all benchmark fine-tuning data. The $\pi_{0.5} + \mathrm{Ego}$ result is reported as an exploratory additional condition. Standard ACT is not evaluated because its categorical task conditioning cannot specify an unseen task without learning new task-specific parameters. Any success is interpreted only as transfer to this held-out task, not as general open-world capability.

H6: Perturbation recovery
Under matched physical perturbations, ACT and $\pi_{0.5}$ may differ in recovery success even when their nominal success rates are similar. This is interpreted as robustness to the tested interventions, not as a general measure of self-correction. The $\pi_{0.5} + \mathrm{Ego}$ result is reported as an exploratory additional condition.

VI. POLICY MODELS

A. Observations, Actions, and Control

At time t, both policy families receive synchronized RGB observations from the four robot cameras,

$$\mathbf{o}_t = \{\mathbf{o}_t^{\mathrm{head}}, \mathbf{o}_t^{\mathrm{body}}, \mathbf{o}_t^{\mathrm{left\ wrist}}, \mathbf{o}_t^{\mathrm{right\ wrist}}\}.$$

together with the robot configuration $\mathbf{q}_t$. The D415 depth stream is not used as a robot-policy input. $\pi_{0.5}$ is conditioned on the natural-language instruction ℓ. Single-task ACT specialists require no task identifier; the multi-task ACT model receives a learned categorical embedding for the current task ID (T1–T7), providing a simple task-conditioning signal without adding a pretrained language encoder.

The 12-dimensional robot configuration is ordered as

$$\begin{aligned} \mathbf{q}_t &= [\mathbf{q}_t^{L,\mathrm{arm}}, g_t^L, \quad \mathbf{q}_t^{R,\mathrm{arm}}, g_t^R], \\ \mathbf{q}_t^{L,\mathrm{arm}}, \mathbf{q}_t^{R,\mathrm{arm}} &\in \mathbb{R}^5, \\ g_t^L, g_t^R &\in [0,1]. \end{aligned}$$

Only the ten arm-joint components enter the forward- and inverse-kinematics calculations; the two gripper components are independent normalized actuator states.

Both model families predict action chunks with horizon $H = 50$. ACT predicts delta commands for the five pose-controlling joints of each arm together with one absolute normalized gripper target per arm:

$$\pi_\theta^{\mathrm{ACT}}\Bigg([\Delta\mathbf{q}^{L,\mathrm{arm}}, g^L, \quad \Delta\mathbf{q}^{R,\mathrm{arm}}, g^R]_{t:t+49} \;\Big|\; \mathbf{o}_t, \mathbf{q}_t, e_{\mathrm{task}} \Bigg). \tag{1}$$

For $\pi_{0.5}$, robot and human motion are expressed in a shared Cartesian task space. For arm $a \in \{L, R\}$, forward kinematics convert the measured robot joint configuration into an end-effector pose,

$$\mathbf{T}_t^a = \mathrm{FK}(\mathbf{q}_t^{a,\mathrm{arm}}), \tag{2}$$

and the future poses in an action chunk are represented relative to the current pose:

$$\begin{aligned} \Delta\mathbf{T}_{t,h}^a &= (\mathbf{T}_t^a)^{-1}\,\mathbf{T}_{t+h}^a, \\ h &= 0, \ldots, H-1. \end{aligned} \tag{3}$$

The egocentric human hand trajectories are converted to an analogous relative pose representation after compensating for head-camera motion. This provides a common geometric supervision signal without requiring a fictitious mapping between human arm joints and robot joint angles.

At deployment, the policy predicts what each end effector should do, while the inverse-kinematics controller determines how the robot joints should realize that motion. Each predicted relative pose is composed with the current end-effector pose to obtain a desired Cartesian target. The inverse-kinematics solver is initialized from the current joint configuration and enforces joint limits while preferring temporally continuous solutions. Because each arm has five joints that affect its end-effector pose, an arbitrary 6-DoF target is not always exactly reachable. The controller therefore uses a weighted Cartesian objective that prioritizes translation and the task-relevant orientation components, projecting the desired pose onto the arm's reachable task space.

The gripper is represented separately from the Cartesian pose. For each arm, the raw servo position s is normalized to an absolute gripper command $g \in [0, 1]$:

$$g = \mathrm{clip}\left(\frac{s - s_{\mathrm{open}}}{s_{\mathrm{closed}} - s_{\mathrm{open}}}, 0, 1 \right), \tag{4}$$

where s_{open} and s_{closed} are the calibrated servo positions for that gripper, $g = 0$ denotes fully open, and $g = 1$ denotes fully closed. The value g is a normalized actuator state rather than an additional end-effector pose coordinate. Thus, each $\pi_{0.5}$ action contains relative Cartesian targets for the left and right end effectors together with one absolute gripper command per arm. Robot demonstrations supervise all action channels. Human batches supervise only the aligned Cartesian channels because the egocentric recordings do not provide reliable robot-gripper targets; their gripper losses are masked rather than assigned artificial values.

This action-space difference is one reason that direct ACT–$\pi_{0.5}$ comparisons are treated as system-level rather than architecture-only comparisons.

At deployment, the controller operates at $50\,\mathrm{Hz}$, and the first $K = 25$ actions from each predicted chunk are executed before the policy replans. This corresponds to $0.5\,\mathrm{s}$ of control between successive policy predictions, following the inference protocol used by the original π_0 policy [5]. We do not use temporal ensembling in the main experiments because it reduced performance in preliminary tests.

B. Behavior Cloning and Action Chunking

Behavior cloning fits a policy to demonstration pairs by supervised learning. It is simple and stable, but its training distribution is generated by the demonstrator and generally contains mostly correct or successful behavior. Consequently, the policy receives limited supervision for states caused by mistakes or unexpected perturbations. During autonomous execution, small errors can therefore move the robot into unseen states and compound over time. Recovery demonstrations, trajectory perturbations, interactive data aggregation, reinforcement learning, and recovery-focused fine-tuning can all reduce this mismatch.

ACT uses a Transformer to predict action chunks and a conditional variational autoencoder (CVAE) latent during training [2]. The latent variable helps represent multimodal demonstrations by capturing variations between multiple valid action strategies. Chunking captures short-term motion intent, reduces the effective decision horizon, and can smooth control through temporal ensembling. ACT is comparatively compact and can be trained as either a specialist or, in our multi-task condition, a task-ID-conditioned shared policy.

C. π_0 Family: Pretrained VLA with Flow Matching

The π_0 family comprises vision-language-action (VLA) models that map visual observations, language instructions, and robot

state to continuous actions [5], [7]. The specific policy instantiated in all experiments below is $\pi_{0.5}$.

The complete $\pi_{0.5}$ policy consists of a PaliGemma vision-language backbone and a robotics-specific action expert. The backbone is initialized from large-scale image–text pretraining [16], but the cross-embodiment prior does not come from this Internet pretraining alone. Before adaptation to our robot, the complete $\pi_{0.5}$ checkpoint has been jointly trained on heterogeneous data from multiple robot platforms, tasks, camera configurations, and action spaces. Cross-embodiment knowledge therefore enters through the shared VLA checkpoint—including both its robot-adapted representations and action expert—rather than through a newly initialized robot-specific output head.

The π_0-family action expert generates continuous action chunks using conditional flow matching. Starting from a noisy action sample $\mathbf{x}_\tau$ at flow time $\tau \in [0, 1]$, the model learns a velocity field v_θ that transports samples from a simple noise distribution toward the demonstrated action distribution:

$$\frac{d\mathbf{x}_\tau}{d\tau} = v_\theta\Big(\mathbf{x}_\tau, \tau \;\Big|\; \mathbf{o}_t, \mathbf{q}_t, \ell\Big). \tag{5}$$

Because flow matching models a distribution over action sequences, it can represent multiple valid manipulation strategies rather than averaging them into a single action. Its main drawbacks are greater compute and memory requirements.

a) Parameter-efficient adaptation.: To limit training memory while preserving the pretrained cross-embodiment representations, we initialize every VLA condition from the same complete $\pi_{0.5}$ base checkpoint and freeze the original weights of both the vision-language backbone and the pretrained action expert. Trainable low-rank adaptation (LoRA) modules are inserted into the attention layers of both components. The proprioception projection, action-input projection, and action-output projection remain fully trainable. This fixed parameterization is used for every VLA condition; the action expert is never fully fine-tuned.

Restricting updates to LoRA modules and the robotics-specific projection layers reduces the risk of catastrophic forgetting, in which adaptation to the custom robot could overwrite visual, semantic, or manipulation representations learned during pretraining. Parameter-efficient adaptation limits rather than eliminates this risk; the retained capabilities are therefore treated as a pretrained prior rather than assumed to remain perfectly unchanged.

The robot-only $\pi_{0.5}$ and $\pi_{0.5} + \mathrm{Ego}$ conditions use exactly the same trainable parameter set, LoRA rank, optimizer, learning-rate schedule, initialization, and checkpoint-selection procedure. The same backbone and action-expert LoRA modules, together with the same trainable projection layers, are shared across robot and human batches. Consequently, the egocentric ablation changes the training data mixture without introducing additional trainable capacity.

Both policy families predict action chunks, but they differ substantially in scale, initialization, task conditioning, output parameterization, and adaptation procedure. The within-ACT and within-$\pi_{0.5}$ ablations therefore provide the cleaner evidence about simulation and task diversity; direct ACT–$\pi_{0.5}$ differences are reported only descriptively.

VII. Data

A. Target-Robot Behavior-Cloning Data

Real-robot demonstrations are collected on the custom bimanual robot through leader–follower teleoperation. The fixed head D415 RGB stream, fixed body RGB stream, and both wrist RGB streams are recorded synchronously at 30 Hz. The D415 depth stream is disabled for the robot-policy dataset and is not used by ACT or $\pi_{0.5}$ during robot training or deployment. Joint positions and gripper states are recorded at the same time. For ACT, these recorded robot states directly define the action targets. For $\pi_{0.5}$, forward kinematics convert the robot trajectories to relative dual-arm Cartesian end-effector trajectories, with gripper commands retained as robot-specific action channels. Only demonstrations that ultimately complete the task are retained, including those that contain successful recovery behaviors.

B. Simulation Data

A MuJoCo digital twin is constructed from the robot's CAD and kinematic model. Simulated demonstrations are generated by defining task-specific Cartesian waypoints and using inverse kinematics to convert them into joint trajectories. The simulated robot contains the same four observation cameras as the real platform: fixed head, fixed body, left wrist, and right wrist. Their nominal poses are matched to the physical camera mounts and remain fixed in the main experiments. Object poses, visual appearance, physical parameters, and control noise are randomized to improve sim-to-real transfer. The randomized parameters and their sampling ranges are reported in Appendix A. The same simulated trajectory can therefore provide joint-space targets for ACT and Cartesian end-effector targets for $\pi_{0.5}$.

C. Egocentric Human Data

The egocentric dataset contains first-person RGB-D recordings of humans performing tasks aligned with the robot benchmark. For this data collection only, the Intel RealSense D415 depth stream is enabled. The D415 is worn in a head-mounted configuration and records synchronized RGB and depth. MediaPipe Hands [18] detects 21 hand landmarks in the RGB image. For each visible landmark, the aligned depth measurement and calibrated camera intrinsics are used to back-project the image

TABLE II
BALANCED COMPOSITION OF THE AVAILABLE TRAINING DATASETS.

Source	Tasks	Episodes/task	Total episodes	Hours
Real-robot BC	7	60	420	10.5
Simulation	7	500	3,500	58.4
Egocentric human	7	100	700	17.5

coordinate to a metric 3D point. Robust local depth statistics and temporal filtering are used to reduce missing or noisy depth values.

A compact human "end-effector" frame is then constructed from the 3D hand geometry. The palm position is derived from the wrist and metacarpophalangeal landmarks, while palm orientation is estimated from orthogonal axes spanning the wrist, index-base, and little-finger-base landmarks. The resulting 6-DoF hand poses are tracked over time and converted to relative pose trajectories. Camera ego-motion from the head-mounted recording is compensated using the RGB-D sequence so that the pseudo-actions primarily describe hand motion rather than head motion. The exact filtering and camera-motion parameters are reported with the released preprocessing code.

Human videos do not contain robot joint positions or robot joint actions. We therefore do not fabricate joint labels. Instead, following the cross-embodiment co-training idea of Kareer et al. [10], we represent human low-level actions as relative 6-DoF hand end-effector trajectories and represent $\pi_{0.5}$ robot actions in the analogous relative end-effector space. The human left/right hand tracks supervise the corresponding Cartesian action channels. Human gripper labels are not synthesized; gripper-action loss is masked on human batches and remains supervised only by robot data. Each human trajectory also carries the task language instruction, and short subtask descriptions may be used where available.

During $\pi_{0.5}$ + Ego fine-tuning, human and robot batches are co-trained in the same optimization run. Robot batches retain the ordinary robot action supervision, while human batches train the shared visual-language representation and the aligned end-effector prediction objective. Thus, the human data can contribute manipulation and visual experience without requiring a fictitious mapping from a human hand to the robot's joint angles. This experiment is specific to the VLA condition and is not applied to standard ACT.

D. Dataset Composition

Table II summarizes each training source in episodes and hours, since adjacent video frames are highly correlated and do not represent independent demonstrations.

All retained demonstrations complete their assigned task, although some include successful recovery behaviors. The table reports the available data pool; the main ablations do not simply consume all available samples. Instead, batch sampling and optimization budgets are matched according to the comparison being performed, as described in Sec. VIII.

TABLE III
TRAINING/EVALUATION MATRIX. EGO DATA ARE NOT USED WITH STANDARD ACT BECAUSE THE HUMAN RECORDINGS LACK ACT'S ROBOT JOINT-ACTION TARGETS.

Condition	ACT	$\pi_{0.5}$
T1, real	Yes	Yes
T1, real + sim. (fixed)	Yes	Yes
T1, real + sim. (additive)	Yes	Yes
7 single-task specialists, real + simulation	Yes	Yes
1 seven-task model, real + simulation	Yes	Yes
1 seven-task model + ego	–	Yes
Held-out shape sorting	Yes	Yes
Perturbation evaluation	Yes	Yes

VIII. TRAINING AND EVALUATION PROTOCOL

We retain the default hyperparameters of the official ACT and $\pi_{0.5}$ implementations unless otherwise stated. Within each model family, the optimizer, learning rate, batch size, image preprocessing, action horizon, and checkpoint-selection procedure are kept fixed across the relevant comparisons. Training data are split by complete episodes to avoid frame-level leakage. Validation mean-squared error (MSE) is measured every $[TODO:]$ steps, used to select the checkpoint for each run, and reported as a training diagnostic; the primary reported metric is closed-loop task success.

The experimental matrix is summarized in Table III. Perturbation evaluation reuses the trained multi-task checkpoints and therefore does not require additional policy training.

For the single-task ACT–$\pi_{0.5}$ comparison, both models use the same train/validation split of Task 1 real-robot trajectories and the same evaluation initial conditions. This remains a system-level comparison because model size, pretraining, task conditioning, and action parameterization are not matched.

For the simulation study, we report two comparisons independently for ACT and $\pi_{0.5}$. In the fixed-budget ablation, the real-only and real-plus-simulation conditions use the same total number of optimization steps. Batches assigned to the simulation fraction in the mixed condition are replaced by resampled real-robot batches in the real-only condition; no unique real episode is removed. The real/simulation batch ratio is $[TODO:]$.

In the additive experiment, the real-plus-simulation condition retains the same expected number of real-robot batches as the real-only baseline and adds simulated batches, increasing the total number of optimization steps from $[TODO:]$ to $[TODO:]$. The fixed-budget comparison isolates the substitution of simulated experience for repeated real-data exposure, whereas the additive comparison measures the practical benefit of adding simulation and compute together. Neither comparison

tests simulation against collecting an equal amount of new real-robot data.

For the task-diversity ablation, we train seven independent ACT specialists and one task-ID-conditioned seven-task ACT policy, and separately train seven independent $\pi_{0.5}$ specialists and one seven-task $\pi_{0.5}$ policy. Each specialist sees only the real and simulated data for its task. The multi-task models sample tasks uniformly. If one specialist receives U optimization examples, the corresponding multi-task policy receives approximately $7U$ examples in total so that expected exposure per task is matched. This design allows the specialist-to-multi-task effect to be measured separately for each policy family.

For the egocentric-data ablation, only $\pi_{0.5}$ is considered. The two seven-task VLA conditions use the same total number of optimization steps and the same real-robot batch probability. In the human-data condition, a fraction $[TODO:]$ of batches that would otherwise be sampled from simulation is replaced by aligned egocentric human batches. Human samples supervise the relative hand/end-effector action objective described in Sec. VII; missing human gripper targets are masked rather than imputed.

The held-out-task and controlled-perturbation evaluations follow the same general protocol. Both reuse the selected multi-task checkpoints without additional training, use $[TODO:]$ matched rollout seeds per policy, preserve the same camera observations, controller frequency, action-execution horizon, and task-appropriate timeouts across policies, and report exact successes with 95% confidence intervals. ACT and $\pi_{0.5}$ form the primary descriptive comparison; $\pi_{0.5}$ + Ego is reported as an exploratory additional condition in both evaluations.

For the held-out-task evaluation, we predeclare the instruction "Place the rectangular block in the left container and the cube in the right container" before any policy training or test rollouts. The rectangular block, the shape-based instruction, and the object-to-container assignment are excluded from all real-robot, simulation, egocentric, and validation data used for benchmark fine-tuning. The $\pi_{0.5}$ and $\pi_{0.5}$ + Ego policies receive this natural-language instruction. Because multi-task ACT uses categorical task embeddings rather than language, it receives the existing T3 sorting-task ID; no new embedding is added or trained. The ACT–$\pi_{0.5}$ result is therefore a comparison between complete systems with different task conditioning, not an architecture-only ablation. Since $\pi_{0.5}$ is pretrained, "held out" refers specifically to the benchmark fine-tuning data and does not assert that the task concept was absent from its original pretraining corpus.

Complete success requires both objects to be inside their assigned containers at the end of the rollout, with neither object in the wrong container. Partial success is the fraction of correctly sorted objects, yielding scores of 0, 0.5, or 1. Initial object poses are sampled from the same predeclared set for all three policies.

Checkpoint selection is based only on held-out real-robot episodes. All closed-loop test rollouts use the selected checkpoint and are excluded from model selection and hyperparameter tuning. We report the selected checkpoint step, number of training seeds, total optimization examples, and data mixture for every trained configuration.

IX. Experimental Evaluation and Results

We report direct model-family comparisons separately from within-family ablations. Complete success is the primary nominal metric; partial success is reported for the longer-horizon task, and inference latency is measured on the deployment hardware. Success values are reported with 95% confidence intervals. Simulation and specialist-to-multi-task effects are computed separately for ACT and $\pi_{0.5}$; differences between the two model families are interpreted descriptively rather than as isolated architectural effects.

a) Main comparisons.: The Task 1 ACT–$\pi_{0.5}$ comparison changes complete success by [TODO:] percentage points and median inference latency by [TODO:] ms. We report this only as a comparison between two complete systems and do not attribute the difference to pretraining, architecture, model scale, or action representation in isolation.

Under the fixed-budget protocol, replacing repeated real-data exposure with simulation changes Task 1 success by [TODO:] percentage points for ACT and [TODO:] percentage points for $\pi_{0.5}$. Under the additive protocol, retaining real-data exposure and adding simulation changes success by [TODO:] percentage points for ACT and [TODO:] percentage points for $\pi_{0.5}$, while increasing total optimization examples from [TODO:] to [TODO:]. The first comparison isolates data-source substitution under fixed compute; the second measures the practical simulation-plus-compute effect.

For task diversity, the ACT specialist-to-multi-task difference is [TODO:] percentage points and the $\pi_{0.5}$ difference is [TODO:] percentage points. A positive difference is described as evidence consistent with beneficial cross-task transfer within this benchmark, while a negative difference is described as interference. Finally, replacing a matched fraction of simulation batches with egocentric human data changes $\pi_{0.5}$ multi-task success by [TODO:] percentage points. Any interpretation is restricted to the data mixture, task suite, and compute budget used here.

b) Per-task results.: Table V reports complete success for both model families under specialist and multi-task training, together with the $\pi_{0.5}$ + Ego condition. This breakdown indicates whether average changes are shared across tasks or concentrated on a small subset.

c) Held-out-task generalization.: The held-out shape-sorting instruction is evaluated without fine-tuning under matched initial conditions. Table VI reports complete success for the same three multi-task policies used in the perturbation study.

TABLE IV

CLOSED-LOOP NOMINAL RESULTS. "FIXED" MATCHES TOTAL OPTIMIZATION EXAMPLES; "ADDITIVE" PRESERVES REAL-DATA EXPOSURE AND ADDS SIMULATION UPDATES. "7 SPECIALISTS" DENOTES SEVEN INDEPENDENTLY TRAINED SINGLE-TASK POLICIES EVALUATED ON THEIR CORRESPONDING TASKS AND AVERAGED OVER THE SEVEN-TASK SUITE.

Model	Training condition	Complete success (%)	Partial success (%)	Latency (ms)
ACT	T1, real	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$	T1, real	[TODO:]	[TODO:]	[TODO:]
ACT	T1, real + simulation (fixed budget)	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$	T1, real + simulation (fixed budget)	[TODO:]	[TODO:]	[TODO:]
ACT	T1, real + simulation (additive)	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$	T1, real + simulation (additive)	[TODO:]	[TODO:]	[TODO:]
ACT	7 specialists, real + simulation	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$	7 specialists, real + simulation	[TODO:]	[TODO:]	[TODO:]
ACT	1 seven-task model, real + simulation	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$	1 seven-task model, real + simulation	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5} + \text{Ego}$	1 seven-task model	[TODO:]	[TODO:]	[TODO:]

TABLE V

CLOSED-LOOP COMPLETE SUCCESS RATE (%) BY TASK.

Configuration	T1	T2	T3	T4	T5	T6	T7
ACT specialists + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
ACT multi-task + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$ specialists + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$ multi-task + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5} + \text{Ego}$ multi-task	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

ACT and $\pi_{0.5}$ form the primary descriptive comparison, while $\pi_{0.5} + \text{Ego}$ is an exploratory additional condition. The result is treated as a measurement on one task rather than evidence of broad zero-shot generalization.

d) Controlled perturbation recovery.: Nominal success does not by itself demonstrate recovery: a policy may succeed more often simply because it makes fewer initial errors. We therefore perform a separate perturbation evaluation using matched initial-condition seeds and the trained multi-task checkpoints. Three standardized interventions are applied where relevant: (i) *object displacement*, in which the target cube is moved laterally by $[TODO:]\,\text{cm}$ after the arm begins its approach but before the first grasp; (ii) *post-grasp displacement*, in which a successfully lifted cube is placed back on the workspace at a nearby reachable pose, forcing the policy to reacquire it; and (iii) *drawer reset*, in which an opened drawer is moved $[TODO:]\,\text{cm}$ toward the closed position before the next subtask. Each perturbation is applied at a predefined event rather than at a fixed wall-clock time.

The primary robustness comparison applies identical perturbations to the seven-task ACT and $\pi_{0.5}$ real+simulation policies. We additionally report the $\pi_{0.5} + \text{Ego}$ checkpoint as an exploratory condition. Recovery success is defined as completion of the original instruction after the intervention without human assistance; the recovery gap is the difference from the matched unperturbed completion rate.

e) Qualitative observations.: The main observed failure modes are missed grasps, incorrect object selection, unstable placement, failed handovers, drawer-interaction errors, and loss of visual contact caused by occlusion. We record representative recovery and non-recovery cases from the controlled perturbation suite, but use the quantitative perturbation metric rather than isolated examples to support claims about recovery. Validation MSE is used for checkpoint selection and reported as a training diagnostic, but it is not interpreted as a direct measure of closed-loop task completion.

X. FUTURE WORK

Future experiments should test additional robot embodiments, workspaces, objects, and instructions, and should include learning curves over real-robot, simulation, and egocentric-data scale. An initialization- and action-space- matched control would help separate the effect of VLA pretraining from the other differences between ACT and $\pi_{0.5}$. Additional data-source controls —for example, replacing the human-data fraction with an equal amount of new simulated or real-robot data—would further clarify whether any observed gain is specific to human cross-embodiment experience or to additional distributional diversity.

The existing MuJoCo environments also make reinforcement-learning fine-tuning a natural extension. Starting from the imitation-learned ACT or $\pi_{0.5}$ checkpoints, simulated rollouts could expose the policy to off-demonstration states and optimize task completion or recovery objectives directly. Such an experiment would need to control reward design, exploration,

TABLE VI

PERFORMANCE ON THE SHAPE-CONDITIONED SORTING TASK HELD OUT FROM BENCHMARK FINE-TUNING. ACT AND $\pi_{0.5}$ FORM THE PRIMARY DESCRIPTIVE COMPARISON; $\pi_{0.5}$ + Ego IS EXPLORATORY.

Policy	Successes/rollouts	Success (95% CI)
ACT multi-task + BC + simulation	[TODO:] / [TODO:]	[TODO:]
$\pi_{0.5}$ multi-task + BC + simulation	[TODO:] / [TODO:]	[TODO:]
$\pi_{0.5}$ + Ego multi-task	[TODO:] / [TODO:]	[TODO:]

TABLE VII

CONTROLLED PERTURBATION RECOVERY. VALUES ARE COMPLETE TASK-SUCCESS RATES (%) AFTER INTERVENTION; THE UNPERTURBED COLUMN USES MATCHED ROLLOUT INITIALIZATIONS. ACT AND $\pi_{0.5}$ FORM THE PRIMARY DESCRIPTIVE COMPARISON; $\pi_{0.5}$ + Ego IS EXPLORATORY.

Policy	Unperturbed	Object displacement	Post-grasp displacement	Drawer reset
ACT multi-task + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$ multi-task + BC + simulation	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$\pi_{0.5}$ + Ego multi-task	[TODO:]	[TODO:]	[TODO:]	[TODO:]

and the sim-to-real gap carefully, and is therefore left outside the present ablation study.

A complementary direction is recovery-focused data collection. RaC (Recovery and Correction) fine-tunes an imitation policy on human-in-the-loop intervention trajectories in which an operator first moves the robot back toward a familiar state and then demonstrates a corrective segment [19]. This is particularly relevant to the present platform because the perturbation suite already identifies missed grasps, displaced objects, and drawer-state errors that can be converted into targeted recovery examples. Combining the present nominal demonstrations with RaC-style recovery and correction data is therefore a concrete future experiment for improving closed-loop robustness on Task 7 and the shorter manipulation tasks.

A hierarchical system combining slower semantic planning with a compact, high-frequency controller is another direction, but is outside the scope of the present experiments.

XI. LIMITATIONS

The study is intentionally small and its conclusions should be read within that scope. First, all physical experiments use one custom robot, one workspace, the same four camera placements, seven trained manipulation tasks, and one held-out shape-sorting task. Performance on a single held-out instruction does not establish broad zero-shot task generalization, open-world robustness, or transfer to a second robot embodiment. The term "generalist" is therefore used only in the restricted sense defined in Sec. V.

Second, the ACT–$\pi_{0.5}$ comparison is not an architecture ablation. The two systems differ in pretraining, parameter count, visual-language representation, task conditioning, and low-level action parameterization. The parallel ACT and $\pi_{0.5}$ simulation and task-diversity ablations make it possible to compare within-family effect sizes, but a difference between those effect sizes still does not identify a single causal mechanism such as pretraining alone. The held-out-task comparison retains this system-level interpretation: ACT uses the trained T3 sorting-task ID, whereas $\pi_{0.5}$ receives the new natural-language instruction.

Third, the simulation ablation compares simulated trajectories against repeated exposure to a fixed real dataset; it does not show that simulation is preferable to collecting an equal amount of new real-robot data. The specialist–multi-task comparison tests shared training and parameter sharing as a package and cannot fully separate representation sharing, regularization, task conditioning, or optimization effects.

Fourth, the egocentric-data experiment is specific to $\pi_{0.5}$ and uses pseudo-actions derived from RGB-D hand tracking rather than robot joint labels. MediaPipe landmark error, RealSense depth noise, self-occlusion, camera ego-motion estimation, and the approximation of a human hand by a 6-DoF end-effector frame all introduce supervision noise. Human gripper actions are not directly supervised. A performance difference therefore indicates an effect of the chosen human-data co-training recipe, not proof that human embodiment data will transfer in general. Because standard ACT is not trained with the same human pseudo-action interface, the ego-data condition does not provide an ACT–$\pi_{0.5}$ comparison.

Finally, Tasks 1–6 last roughly 10 s and Task 7 roughly 20 s. Task 7 is longer within this benchmark but remains short compared with many household or industrial long-horizon problems. Confidence intervals over physical rollouts quantify rollout uncertainty, but unless several independently trained seeds are evaluated they do not capture all training-run variance. The controlled perturbations test a small set of predefined disturbances and should not be interpreted as general self-correction capability. These limitations motivate treating the results as evidence about this benchmark and experimental regime rather than broad claims about general-purpose robot learning.

XII. Conclusion

We presented a small-scale study of model choice, simulation, task diversity, and egocentric human supervision on a low-cost fixed-base bimanual robot with 12 actuated DoF: ten arm-joint DoF and two gripper DoF. The experimental design mirrors the simulation and multi-task ablations within both ACT and $\pi_{0.5}$, allowing the effect of these training changes to be measured separately for each policy family while keeping direct ACT–$\pi_{0.5}$ comparisons descriptive. Egocentric RGB-D human data are evaluated only for $\pi_{0.5}$ through a cross-embodiment end-effector representation, and recovery is measured using explicit perturbations rather than inferred from nominal success. The results should be interpreted only for the robot embodiment and benchmark described here. [TODO: Insert the final numerical findings using benchmark-specific language; report the within-ACT and within-$\pi_{0.5}$ simulation and multi-task effect sizes, the ego-data effect, the held-out-task result, and perturbation recovery with uncertainty, without extrapolating beyond the tested tasks, data mixture, and compute budget.]

Reproducibility Statement

The complete software stack for the custom bimanual platform is available at https://github.com/tmeynier/bimanual-models-tasks-data. The repository includes the calibration, demonstration-recording, policy-training, inference, and evaluation code, together with the robot CAD files and MuJoCo simulation environment.

The arms use Feetech STS3215 servomotors, following the widely available LeRobot SO-101 configuration. These motors and their controllers can be purchased from standard online robotics suppliers. The platform consists of the two fixed manipulation arms and their controllers. The released hardware description also includes the fixed-head D415 mount, fixed body-camera mount, and left/right wrist-camera mounts, together with the camera calibration used to reproduce the corresponding MuJoCo viewpoints.

Datasets and trained checkpoints will be released through https://huggingface.co/tmeynier under the following names:

- **Datasets:** `bimanual-real-bc`, `bimanual-mujoco-sim`, and `bimanual-egocentric`.
- **Model checkpoints:** `bimanual-act-bc-1task`, `bimanual-act-bc-sim-1task`, `bimanual-act-bc-sim-task1` through `bimanual-act-bc-sim-task7`, `bimanual-act-bc-sim-7task`, `bimanual-pi05-bc-1task`, `bimanual-pi05-bc-sim-1task`, `bimanual-pi05-bc-sim-task1` through `bimanual-pi05-bc-sim-task7`, `bimanual-pi05-bc-sim-7task`, and `bimanual-pi05-ego-7task`.

These releases include the training configurations, dataset metadata, evaluation seeds, selected checkpoints, and per-rollout results required to reproduce the experiments.

Appendix

Table VIII reports the parameters and sampling ranges used to generate the simulated demonstrations.

TABLE VIII
Domain-randomization parameters used in MuJoCo.

Parameter	Distribution	Range
Object position	Uniform	[TODO:]
Object orientation	Uniform	[TODO:]
Camera pose	Fixed	Matched to real camera mounts
Lighting	Uniform	[TODO:]
Object mass	Uniform	[TODO:]
Contact friction	Uniform	[TODO:]
Action noise	Gaussian	[TODO:]
Control delay	Uniform	[TODO:]

- Exact 12-DoF accounting: five kinematic arm joints plus one normalized gripper actuator per side, including the associated low-level controllers.
- Intrinsics/extrinsics and resolution for the fixed head, fixed body, left-wrist, and right-wrist cameras; confirmation that D415 depth is excluded from robot-policy input; observation history, action representation, chunk horizon, and execution horizon.
- Dataset episodes, hours, task/source composition, failure retention, train/validation/test splitting unit, and licenses/consent.
- Parameter counts, pretrained checkpoint, trainable layers, optimizer, updates, compute, seeds, and checkpoint selection.
- Per-task rollout counts, initial-condition seeds, randomization ranges, success predicates, timeouts, interventions, and safety rules.
- Raw per-rollout outcomes and confidence intervals, not only averages.
- Training seeds and matched optimization budgets for the ACT and $\pi_{0.5}$ simulation and task-diversity ablations.
- Egocentric RGB-D preprocessing details, MediaPipe confidence thresholds, depth filtering, camera-motion compensation, end-effector construction, and masked human action channels.
- Exact perturbation magnitudes, trigger events, applicable tasks, and recovery-success criteria.

1) CAD-to-physical robot overview with the kinematic tree, all axes, and the four real/sim camera viewpoints.
2) Dataset examples from real BC, simulation, and egocentric RGB-D human video, including MediaPipe 3D hand tracks and the derived end-effector frame.
3) One paired rollout per ablation under the same initial-condition seed.
4) Recovery-suite rollouts showing object displacement and grasp failure.

5) Failure taxonomy and latency distribution on deployment hardware.

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